In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')
```

Load the Titanic dataset

In [2]:
```
df=sns.load_dataset('titanic')
df.head()
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Display first five rows

In [3]:
```
df.head()
```

Out[3]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Display last five rows

In [4]:
```
df.tail()
```

Out[4]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
886	0	2	male	27.0	0	0	13.00	S	Second	man	True	NaN	Southampton	no	True
887	1	1	female	19.0	0	0	30.00	S	First	woman	False	B	Southampton	yes	True
888	0	3	female	NaN	1	2	23.45	S	Third	woman	False	NaN	Southampton	no	False
889	1	1	male	26.0	0	0	30.00	C	First	man	True	C	Cherbourg	yes	True
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	NaN	Queenstown	no	True

Dataset shape

In [5]:
```
df.shape
```

Out[5]:
```
(891, 15)
```

Dataset information

In [6]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype   
---  ------       --------------  -----   
 0   survived     891 non-null    int64   
 1   pclass       891 non-null    int64   
 2   sex          891 non-null    str     
 3   age          714 non-null    float64 
 4   sibsp        891 non-null    int64   
 5   parch        891 non-null    int64   
 6   fare         891 non-null    float64 
 7   embarked     889 non-null    str     
 8   class        891 non-null    category
 9   who          891 non-null    str     
 10  adult_male   891 non-null    bool    
 11  deck         203 non-null    category
 12  embark_town  889 non-null    str     
 13  alive        891 non-null    str     
 14  alone        891 non-null    bool    
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 100.4 KB
```

Statistical summary

In [7]:
```
df.describe()
```

Out[7]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Check column names

In [8]:
```
df.columns
```

Out[8]:
```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='str')
```

Check missing values

In [9]:
```
df.isnull().sum()
```

Out[9]:
```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

Data Cleaning

fill missing numerical values

In [10]:
```
df['age']=df['age'].fillna(df['age'].median())
```

fill missing categorical values

In [12]:
```
df['embarked']=df['embarked'].fillna(df['embarked'].mode()[0])
```

Remove duplicate records

In [13]:
```
df.drop_duplicates(inplace=True)
```

Check missing values again

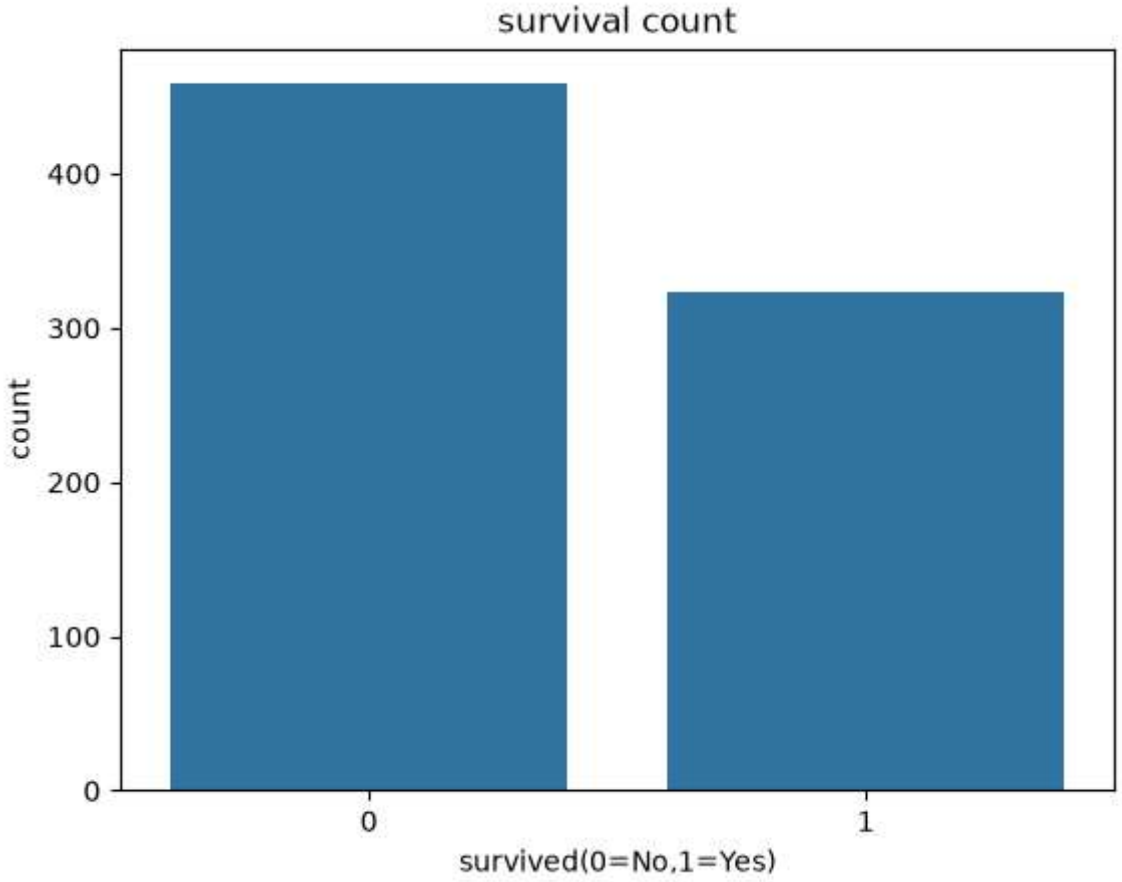
In [14]:
```
df.isnull().sum()
```

Out[14]:
```
survived         0
pclass           0
sex              0
age              0
sibsp            0
parch            0
fare             0
embarked         0
class            0
who              0
adult_male       0
deck           579
embark_town      2
alive            0
alone            0
dtype: int64
```

UNIVARIATE ANALYSIS

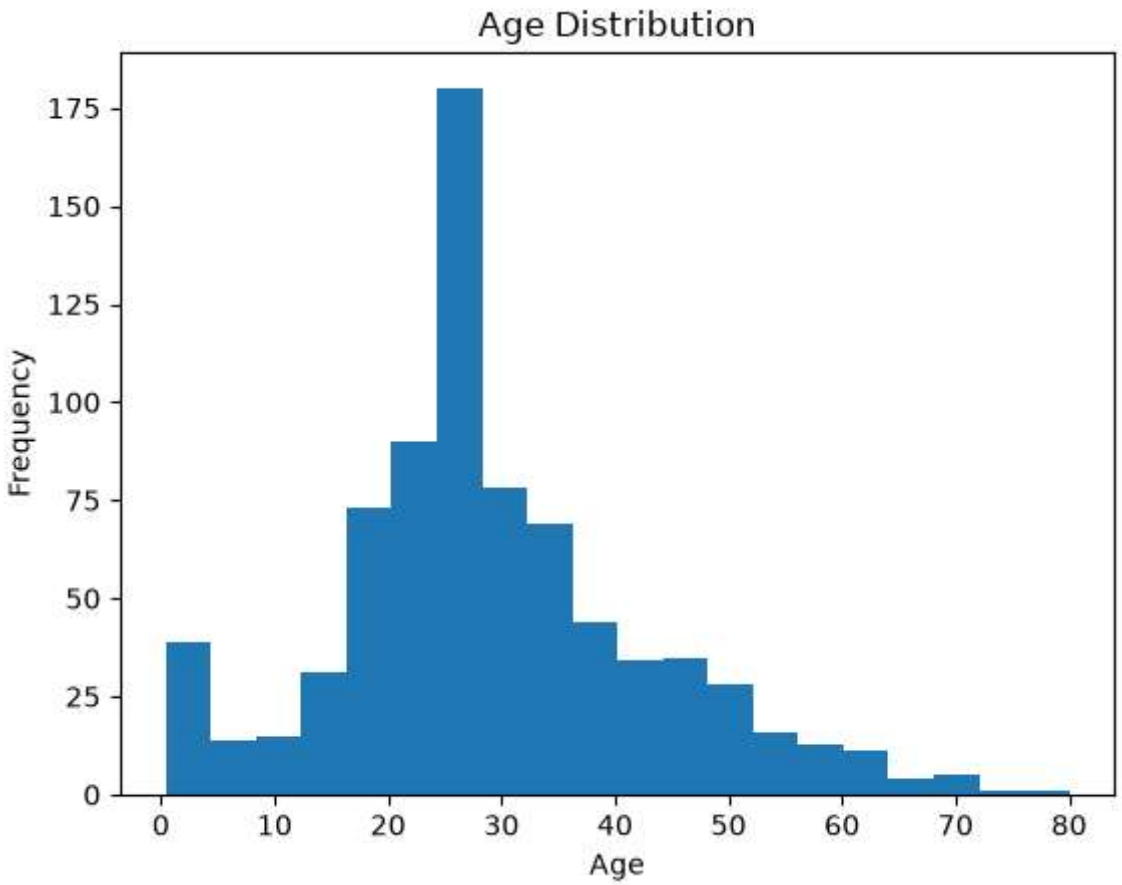
survival count

In [15]:
```
sns.countplot(data=df,x='survived')
plt.title('survival count')
plt.xlabel('survived(0=No,1=Yes)')
plt.ylabel('count')
plt.show()
```

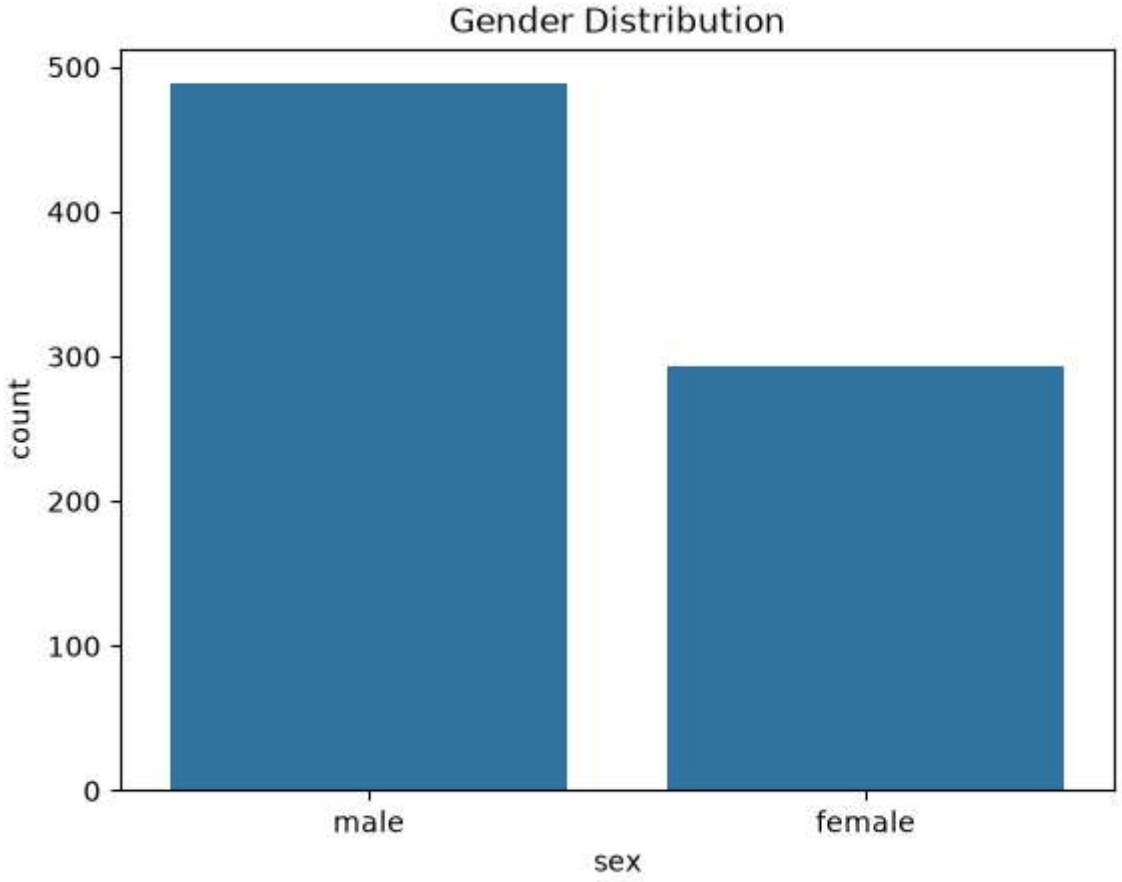


Age Distribution

In [16]:
```
plt.hist(df['age'],bins=20)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



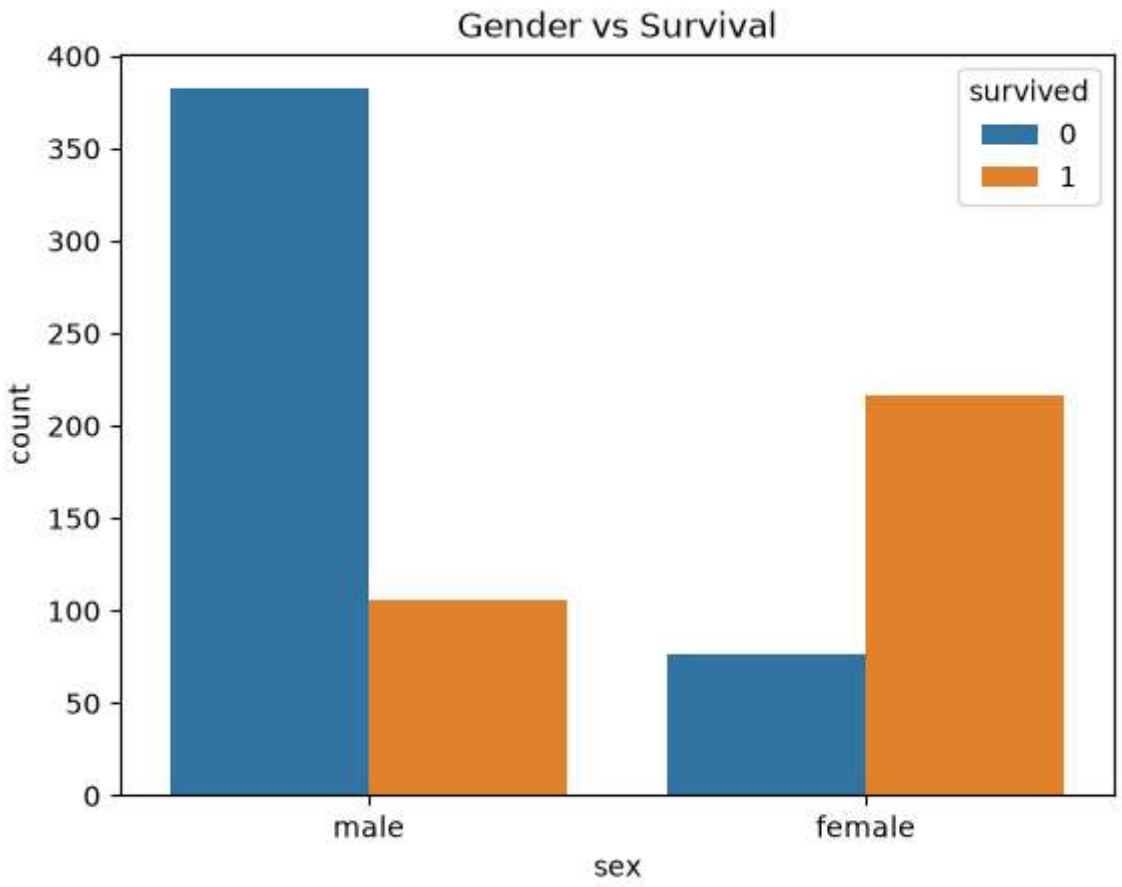
In [17]:
```
sns.countplot(data=df,x='sex')
plt.title('Gender Distribution')
plt.show()
```



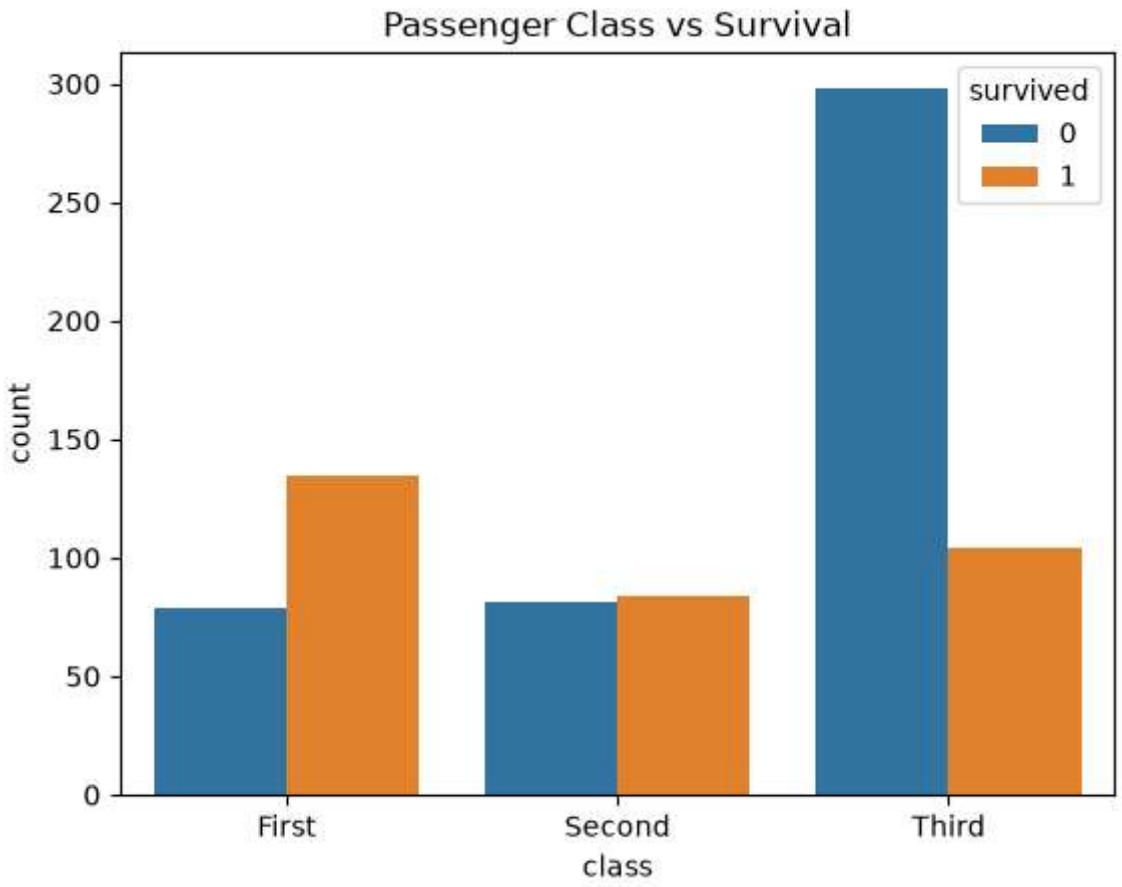
BIVARIATE ANALYSIS

Gender Vs Survival

In [19]:
```
sns.countplot(data=df,x='sex',hue='survived')
plt.title('Gender vs Survival')
plt.show()
```

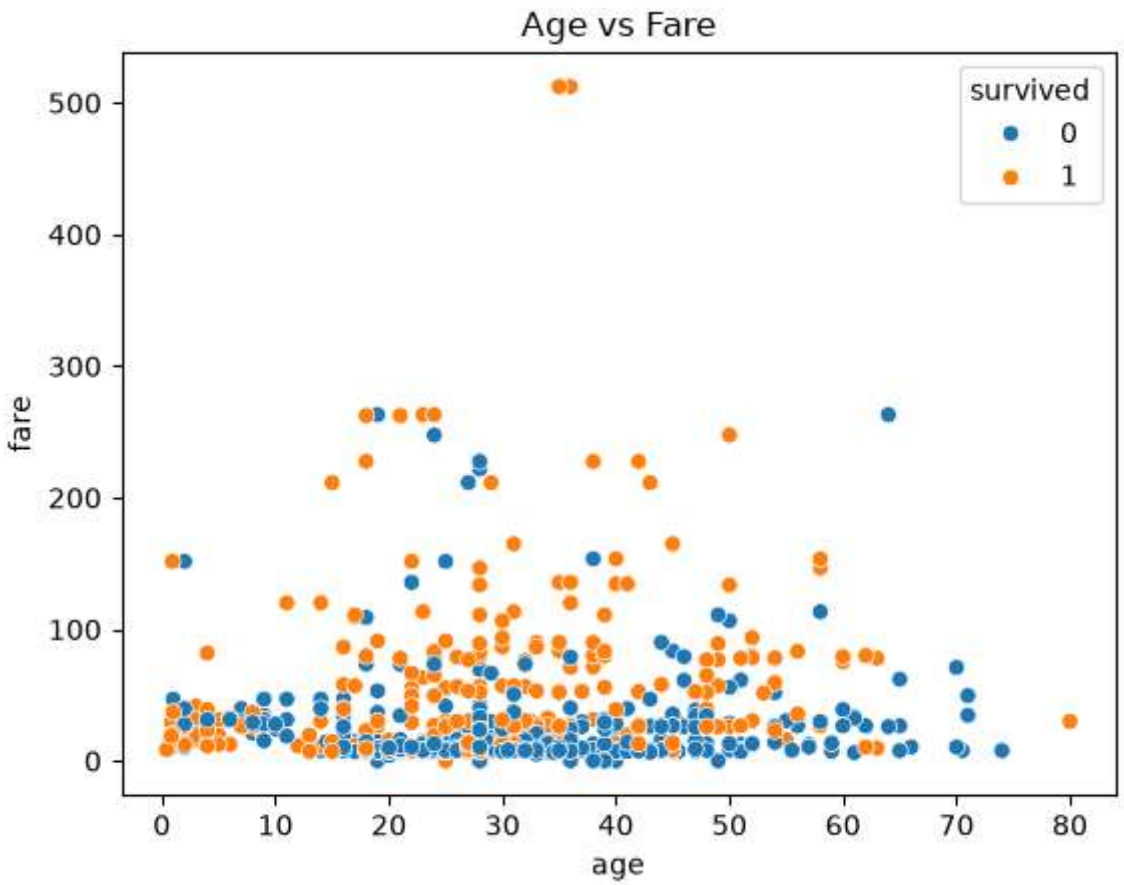


In [21]:
```
sns.countplot(data=df,x='class',hue='survived')
plt.title('Passenger Class vs Survival')
plt.show()
```



Age Vs Fare

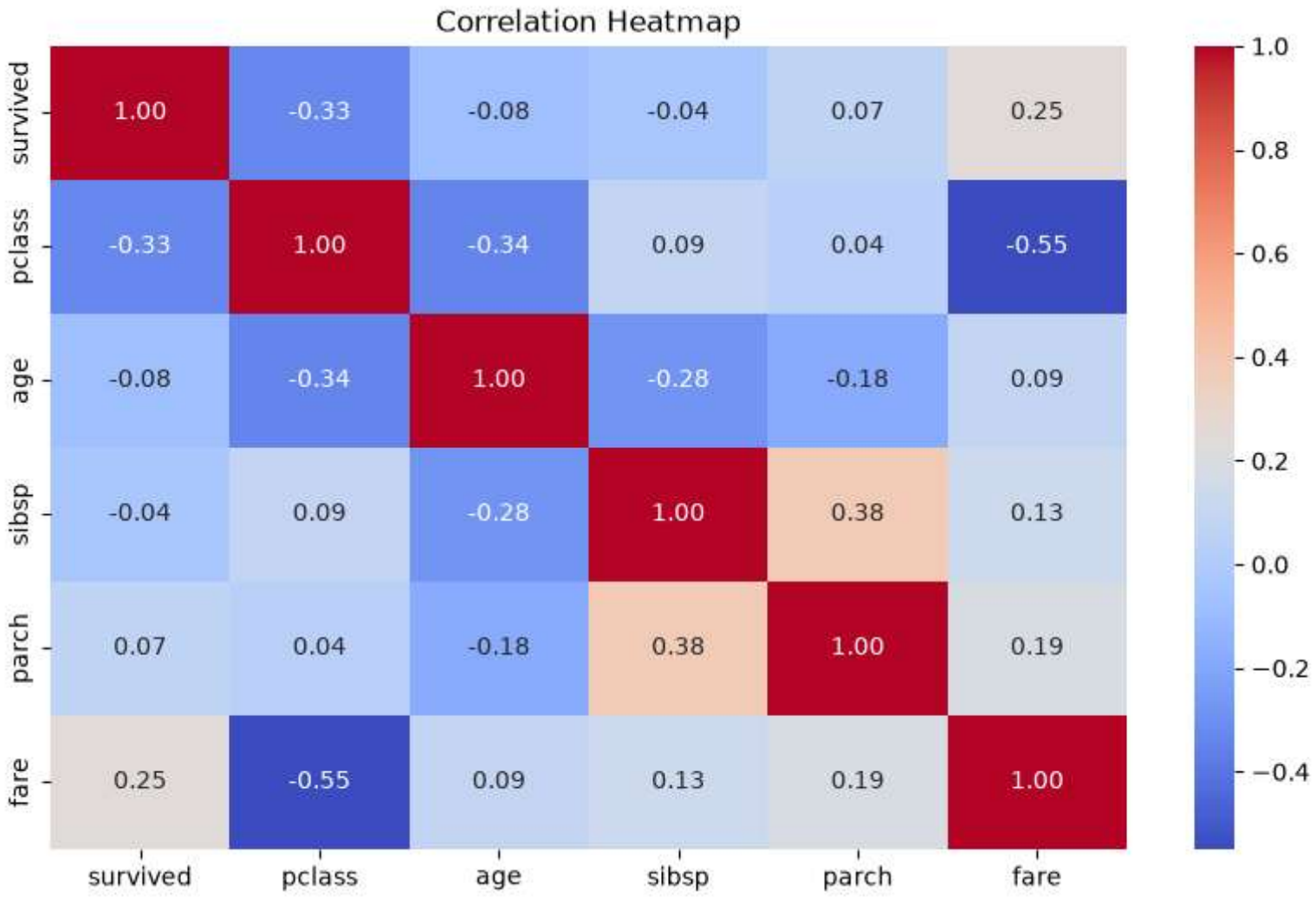
In [22]:
```
sns.scatterplot(data=df,x='age',y='fare',hue='survived')
plt.title('Age vs Fare')
plt.show()
```



Correlation heatmap

In [26]:
```
numeric_df = df.select_dtypes(include=np.number)
plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap='coolwarm',
            fmt='.2f')
```

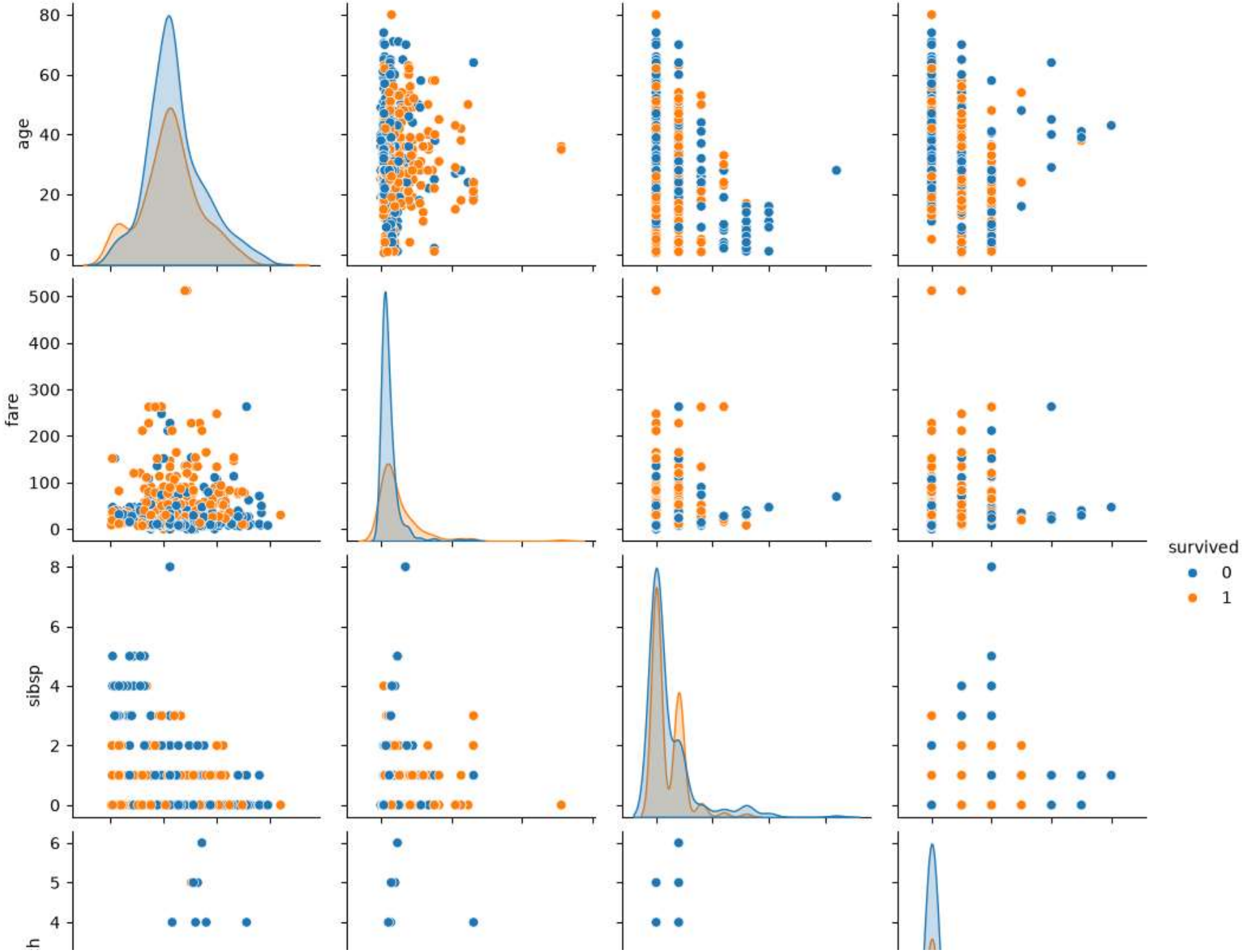
```
plt.title('Correlation Heatmap')
plt.show()
```

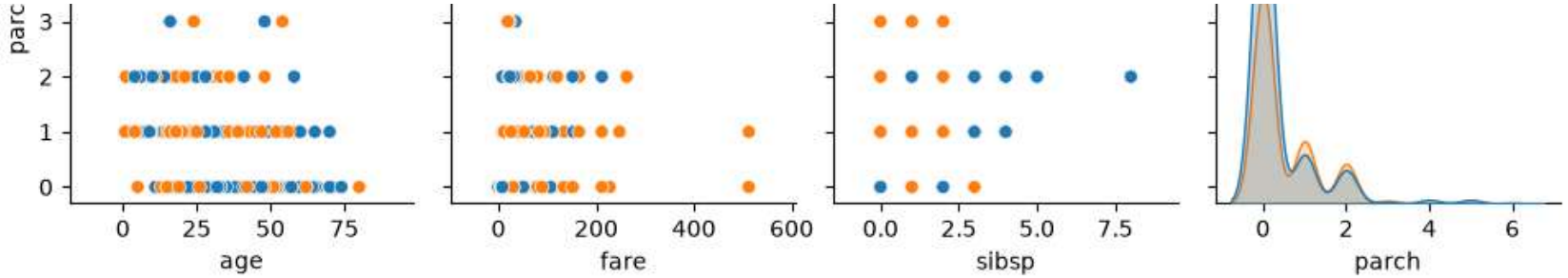


Pairplot

In [27]:
```
sns.pairplot(df[['age','fare','sibsp','parch','survived']],hue='survived')
```

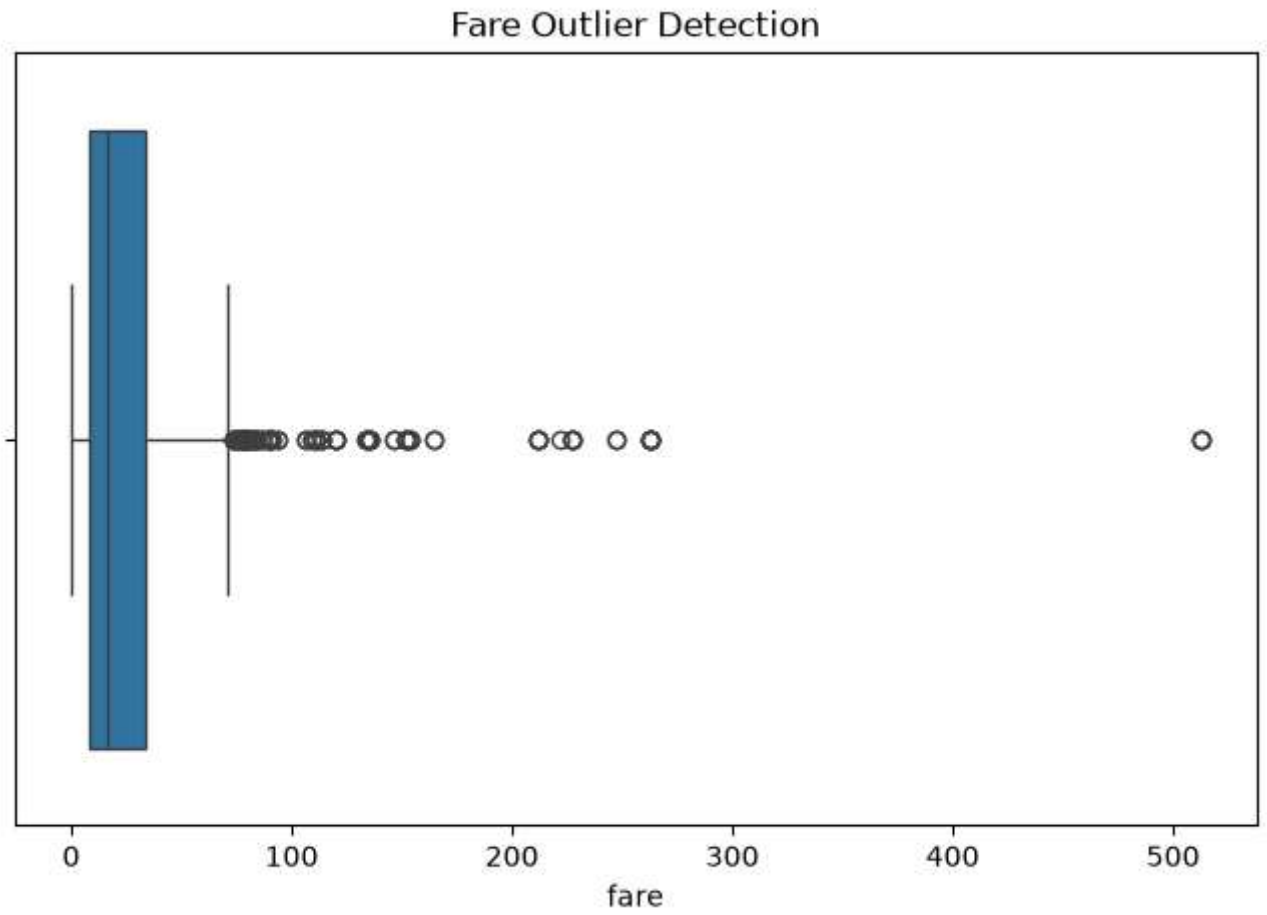
```
plt.show()
```


age
fare
sibsp
survived
0
1



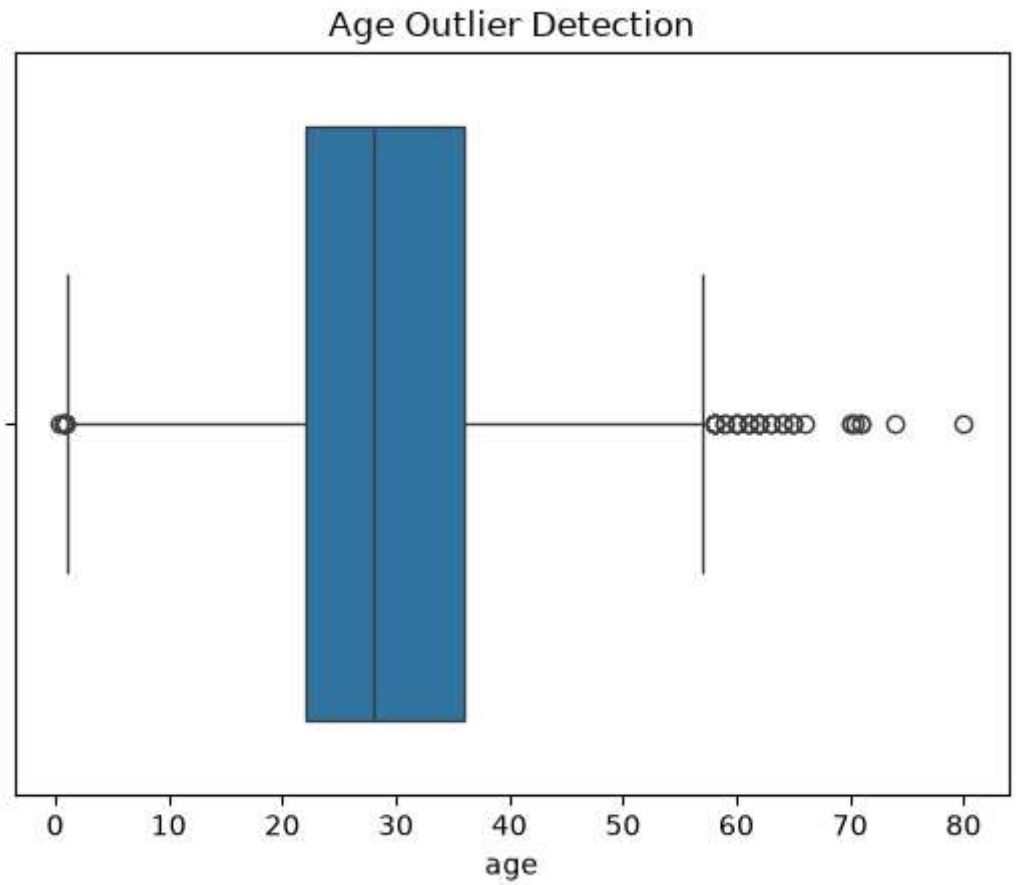
Boxplot-Detect outliers

In [28]:
```
plt.figure(figsize=(8,5))
sns.boxplot(data=df,x='fare')
plt.title('Fare Outlier Detection')
plt.show()
```



Age boxplot

In [29]:
```
sns.boxplot(data=df,x='age')
plt.title('Age Outlier Detection')
plt.show()
```



Value counts

Passenger gender count

In [30]:
```
print(df['sex'].value_counts())
```

```
sex
male      488
female    293
Name: count, dtype: int64
```

Passenger class count

In [31]:
```
print(df['class'].value_counts())
```

```
class
Third     402
First     214
Second    165
Name: count, dtype: int64
```

Survival count

In [32]:
```
print(df['survived'].value_counts())
```

```
survived
0    458
1    323
Name: count, dtype: int64
```

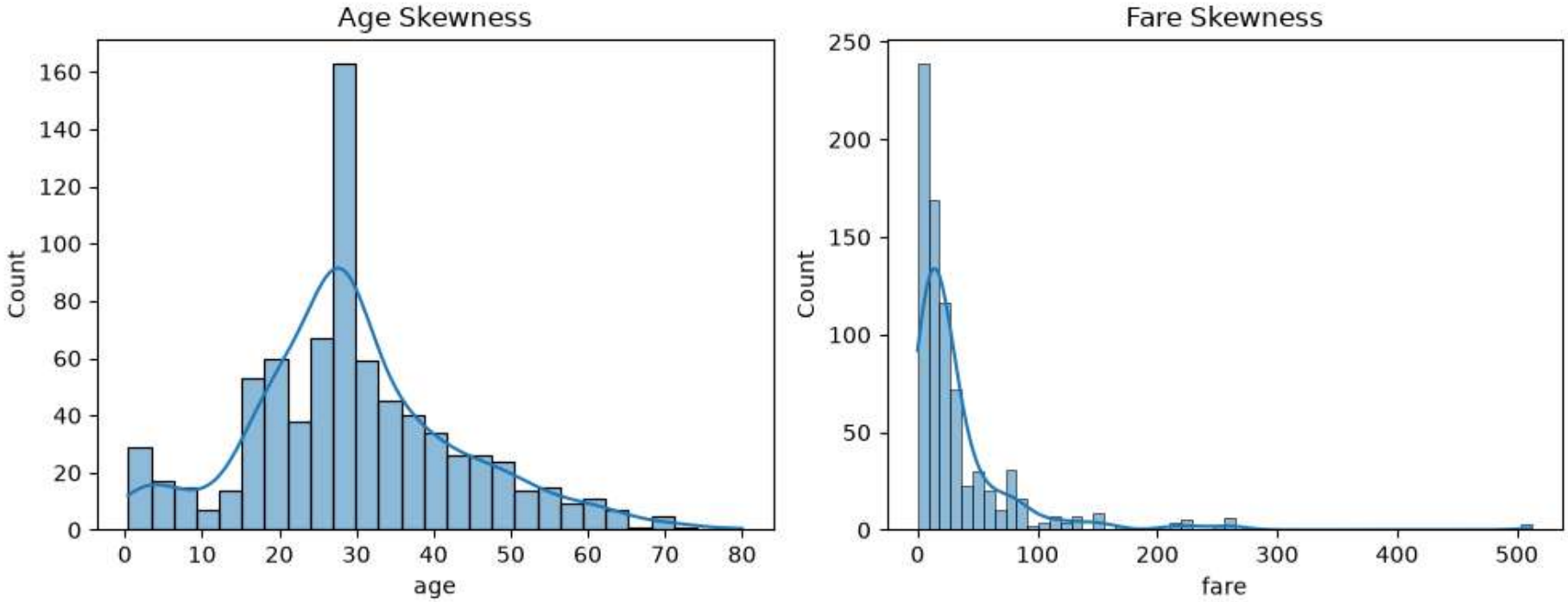
IDENTIFY SKEWNESS

In [33]:
```
print(df[['age','fare']].skew())
```

```
age     0.446558
fare    4.565930
dtype: float64
```

In [34]:
```
plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
sns.histplot(df['age'],kde=True)
plt.title('Age Skewness')
```

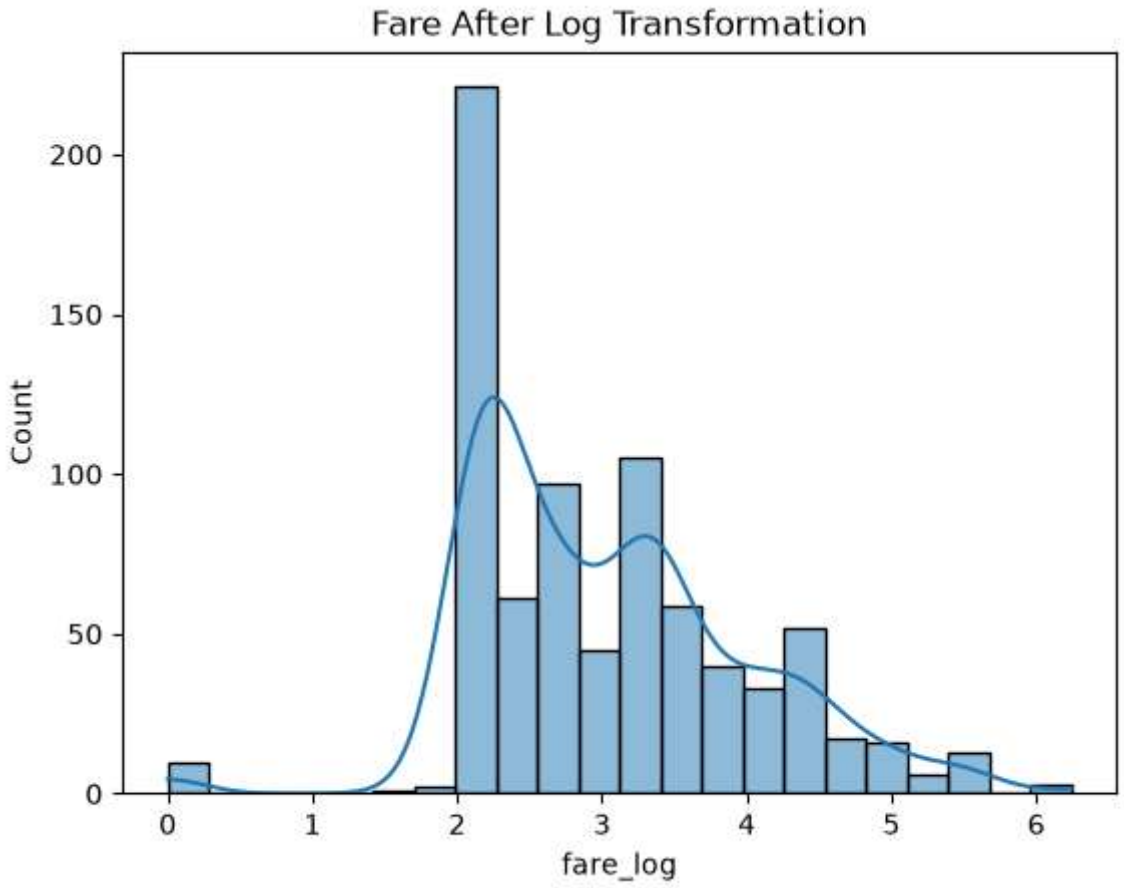
```
plt.subplot(1,2,2)
sns.histplot(df['fare'],kde=True)
plt.title('Fare Skewness')
plt.tight_layout()
plt.show()
```



Handling skewed data

log transformation for fare

In [35]:
```
df['fare_log']=np.log1p(df['fare'])
sns.histplot(df['fare_log'],kde=True)
plt.title('Fare After Log Transformation')
plt.show()
```



Group analysis

Survival rate on gender

In [36]:
```
print(df.groupby('sex')['survived'].mean())
```

```
sex
female    0.740614
male      0.217213
Name: survived, dtype: float64
```

survival rate based on passenger class

In [37]:
```
print(df.groupby('class',observed=True)['survived'].mean())
```

```
class
First     0.630841
Second    0.509091
Third     0.258706
Name: survived, dtype: float64
```

Average fare based on passenger class

In [38]:
```
print(df.groupby('class',observed=True)['fare'].mean())
```

```
class
First     84.487812
Second    21.835404
Third     13.574326
Name: fare, dtype: float64
```

Final statistical analysis

In [40]:
```
print("Total Passengers:",len(df))
print("Survived Passengers:",(df['survived']==1).sum())
print("Not Survived Passengers:",(df['survived']==0).sum())
print("Average Age:",df['age'].mean())
print("Average Fare:",df['fare'].mean())
print("Survival Percentage:",df['survived'].mean()*100)
```

```
Total Passengers: 781
Survived Passengers: 323
Not Survived Passengers: 458
Average Age: 29.62281690140845
Average Fare: 34.75046389244558
Survival Percentage: 41.35723431498079
```